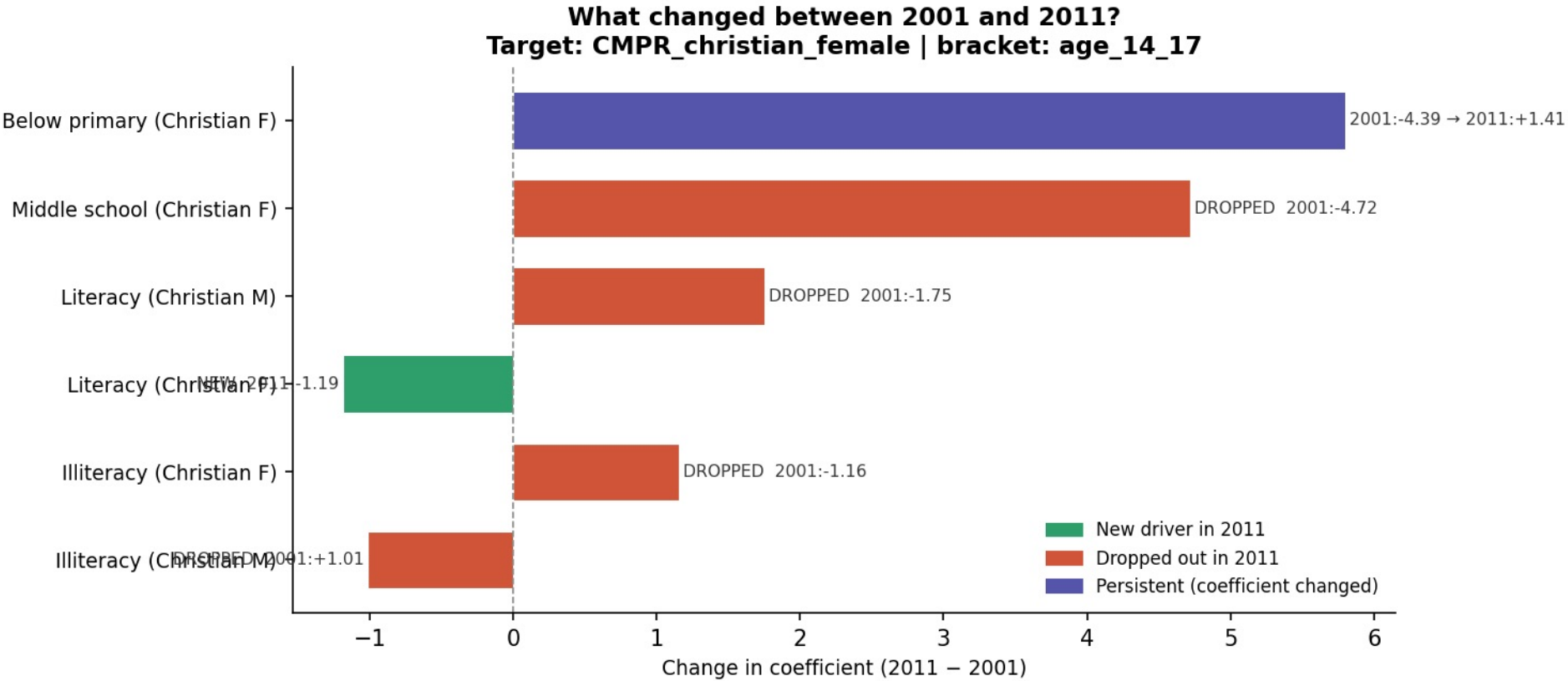
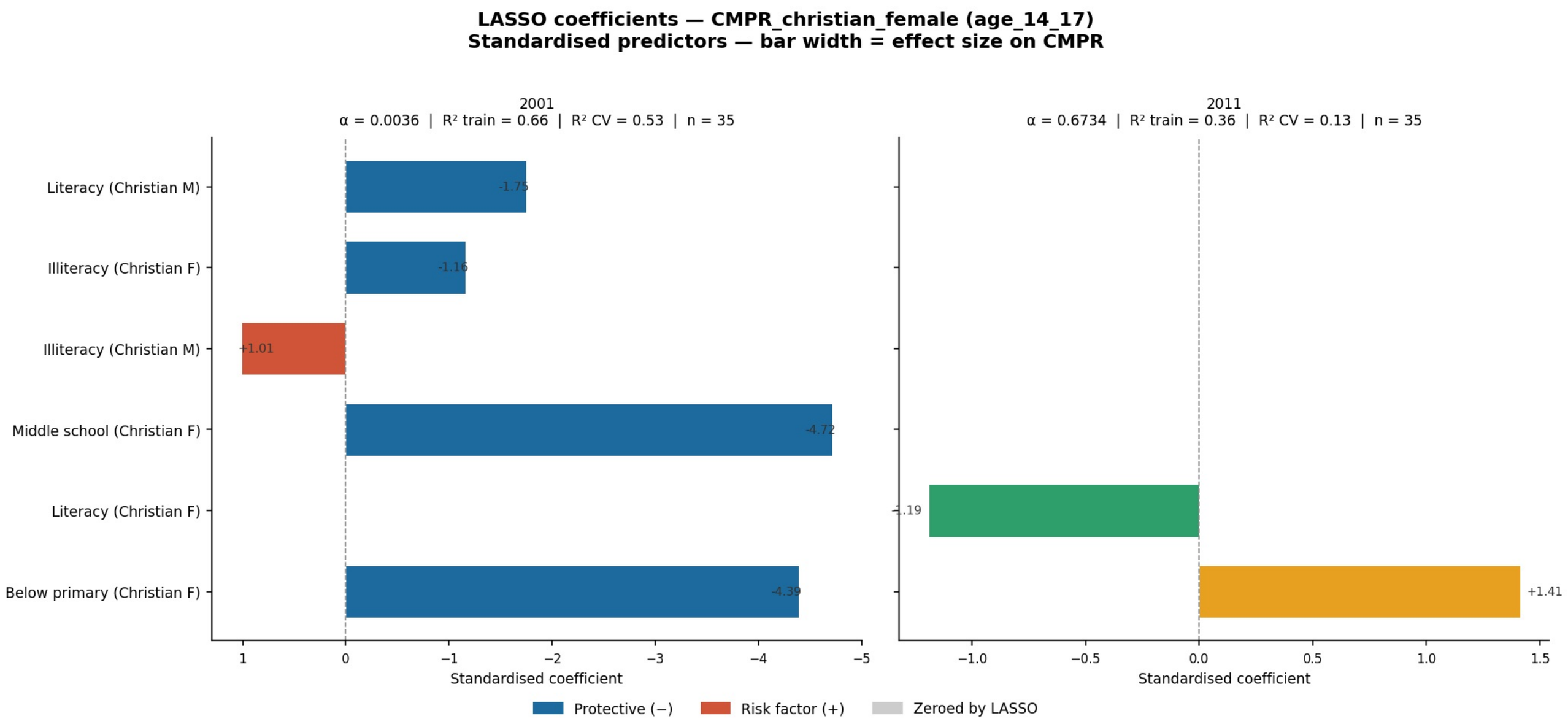
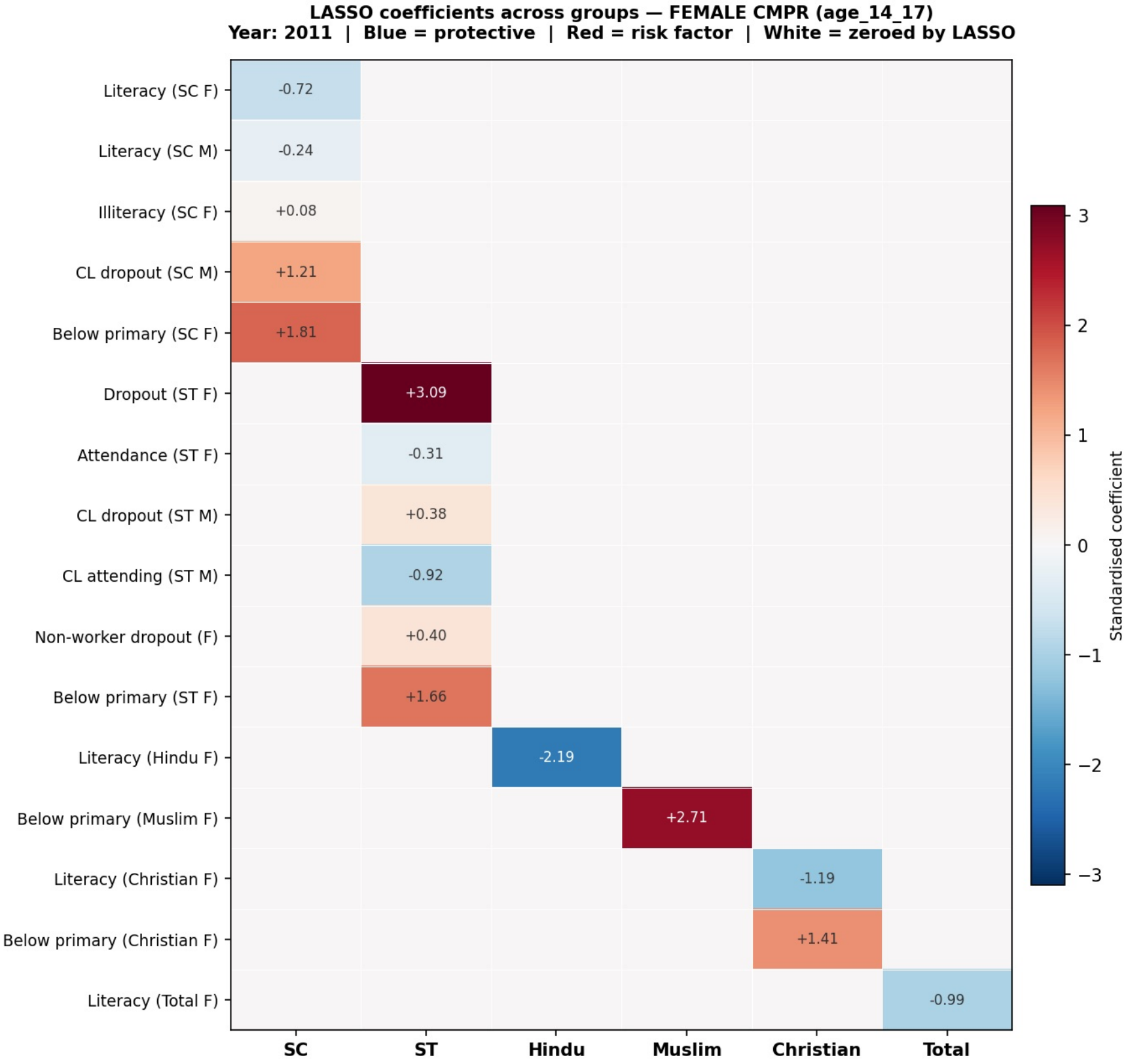
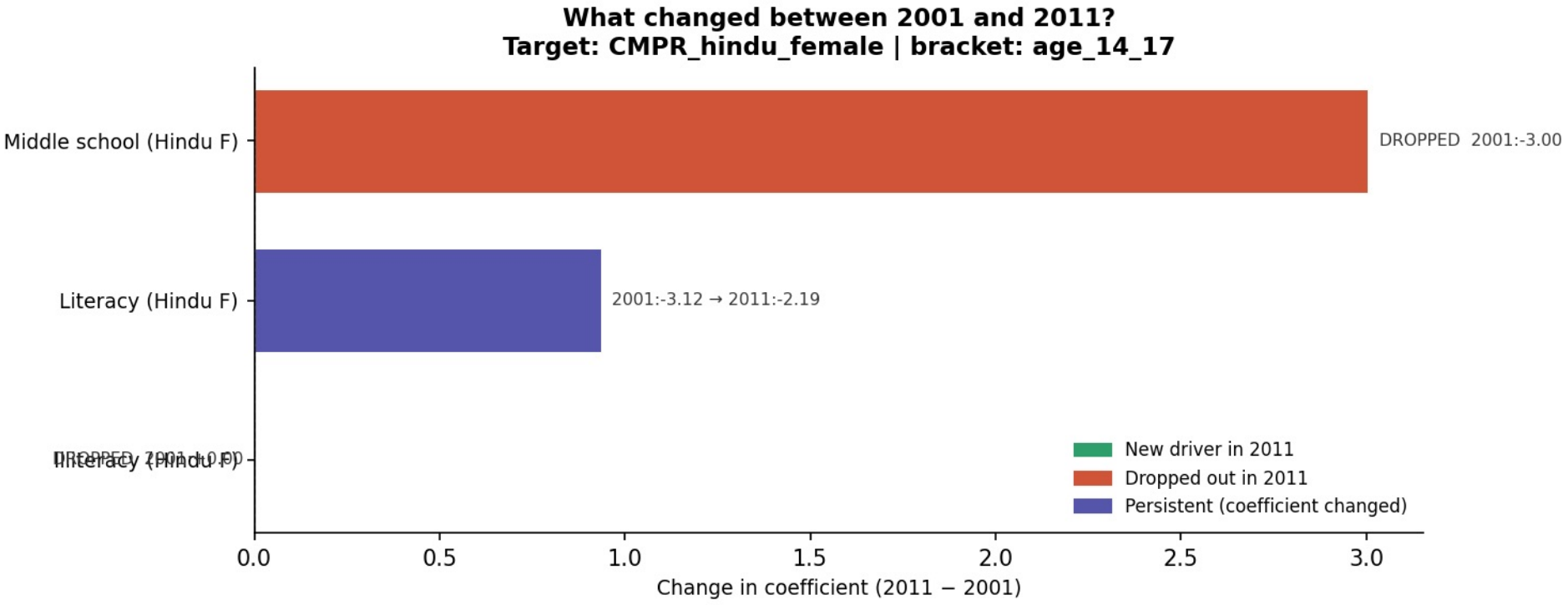
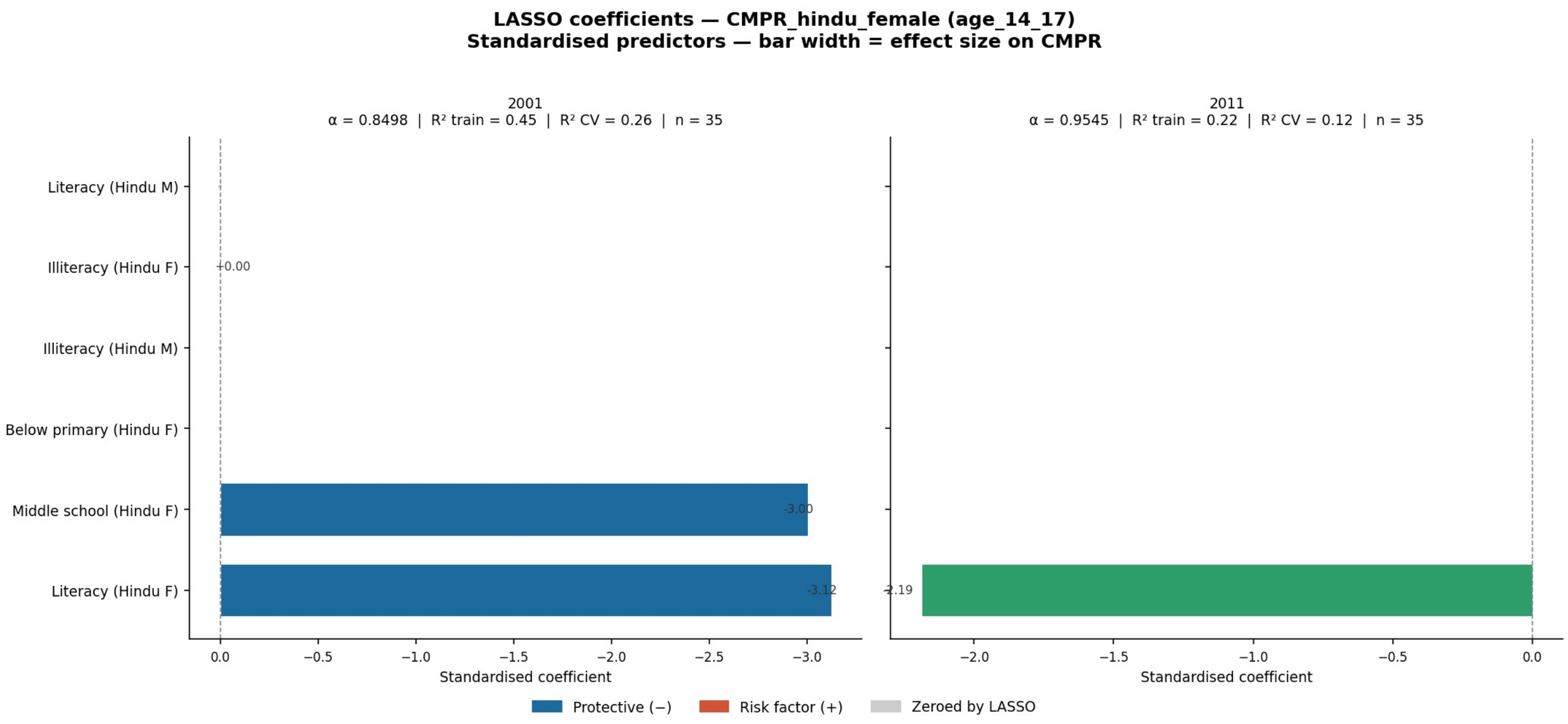
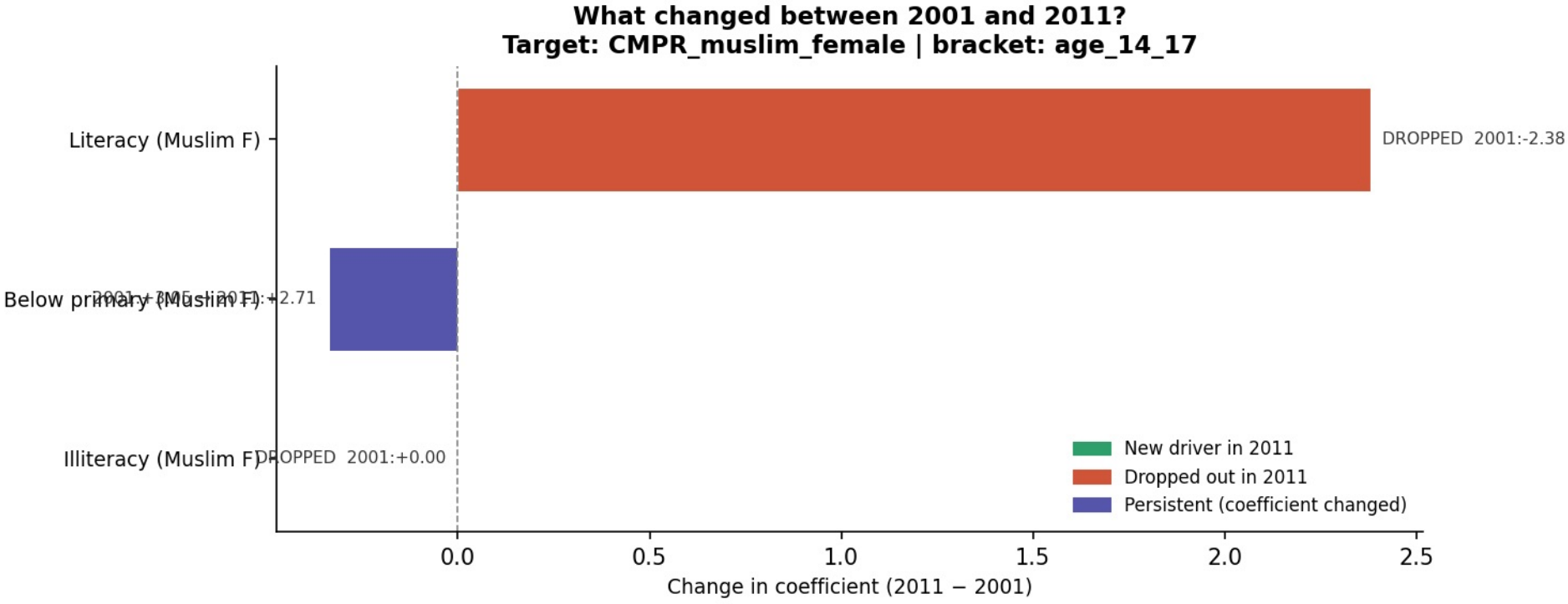

What changed between 2001 and 2011?
Target: CMPR_christian_female | bracket: age_14_17
Below primary (Christian F)
Middle school (Christian F)
Literacy (Christian M)
Literacy (Christian F)
Illiteracy (Christian F)
Illiteracy (Christian M)
2001:-4.39 → 2011:+1.41
DROPPED 2001:-4.72
DROPPED 2001:-1.75
NEW 2011:-1.19
DROPPED 2001:-1.16
DROPPED 2001:+1.01
New driver in 2011
Dropped out in 2011
Persistent (coefficient changed)
−1
0
1
2
3
4
5
6
Change in coefficient (2011 − 2001)

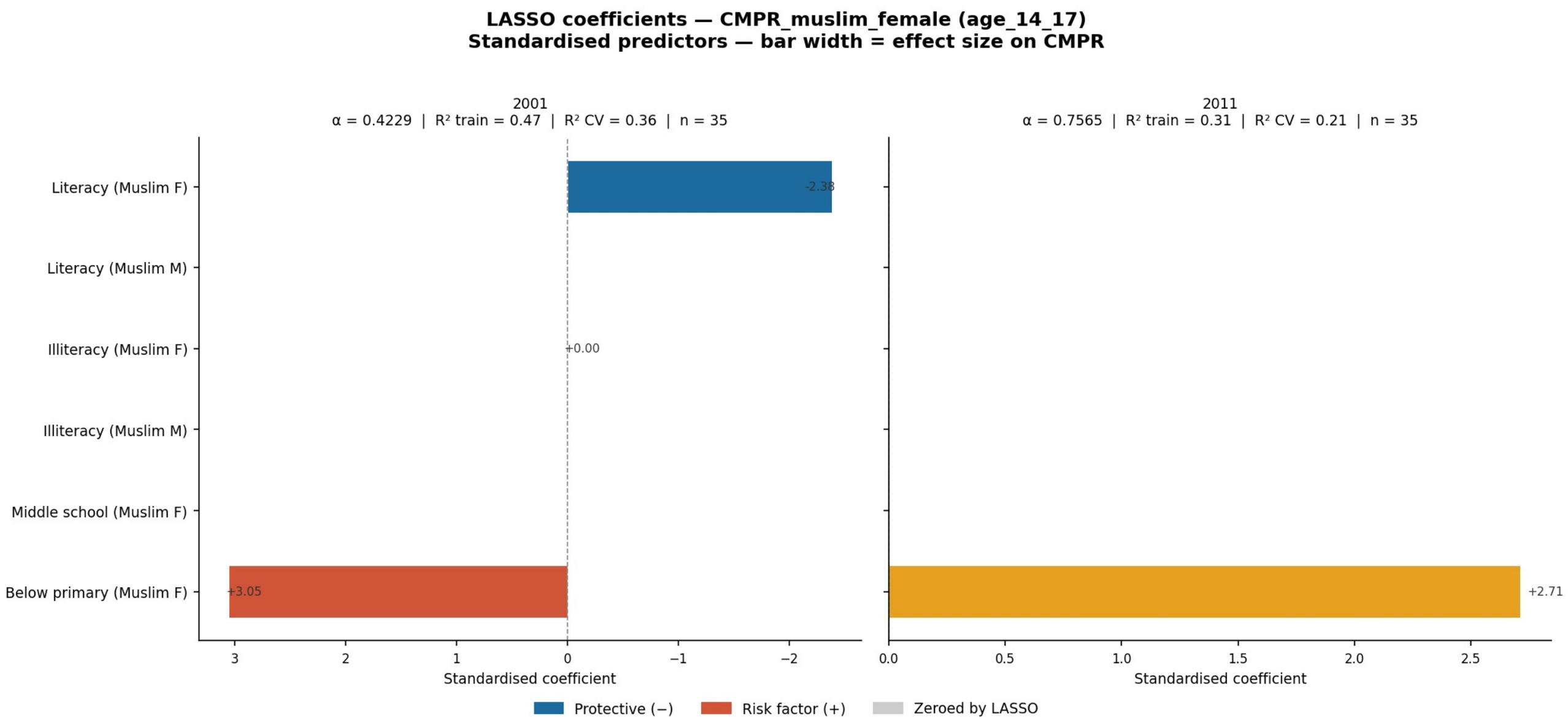

LASSO coefficients — CMPR_christian_female (age_14_17)
Standardised predictors — bar width = effect size on CMPR
2001
α = 0.0036 | R² train = 0.66 | R² CV = 0.53 | n = 35
2011
α = 0.6734 | R² train = 0.36 | R² CV = 0.13 | n = 35
Literacy (Christian M)
Illiteracy (Christian F)
Illiteracy (Christian M)
Middle school (Christian F)
Literacy (Christian F)
Below primary (Christian F)
-1.75
-1.16
+1.01
-4.72
-4.39
-1.19
+1.41
1
0
−1
−2
−3
−4
−5
−1.0
−0.5
0.0
0.5
1.0
1.5
Standardised coefficient
Standardised coefficient
Protective (−)
Risk factor (+)
Zeroed by LASSO

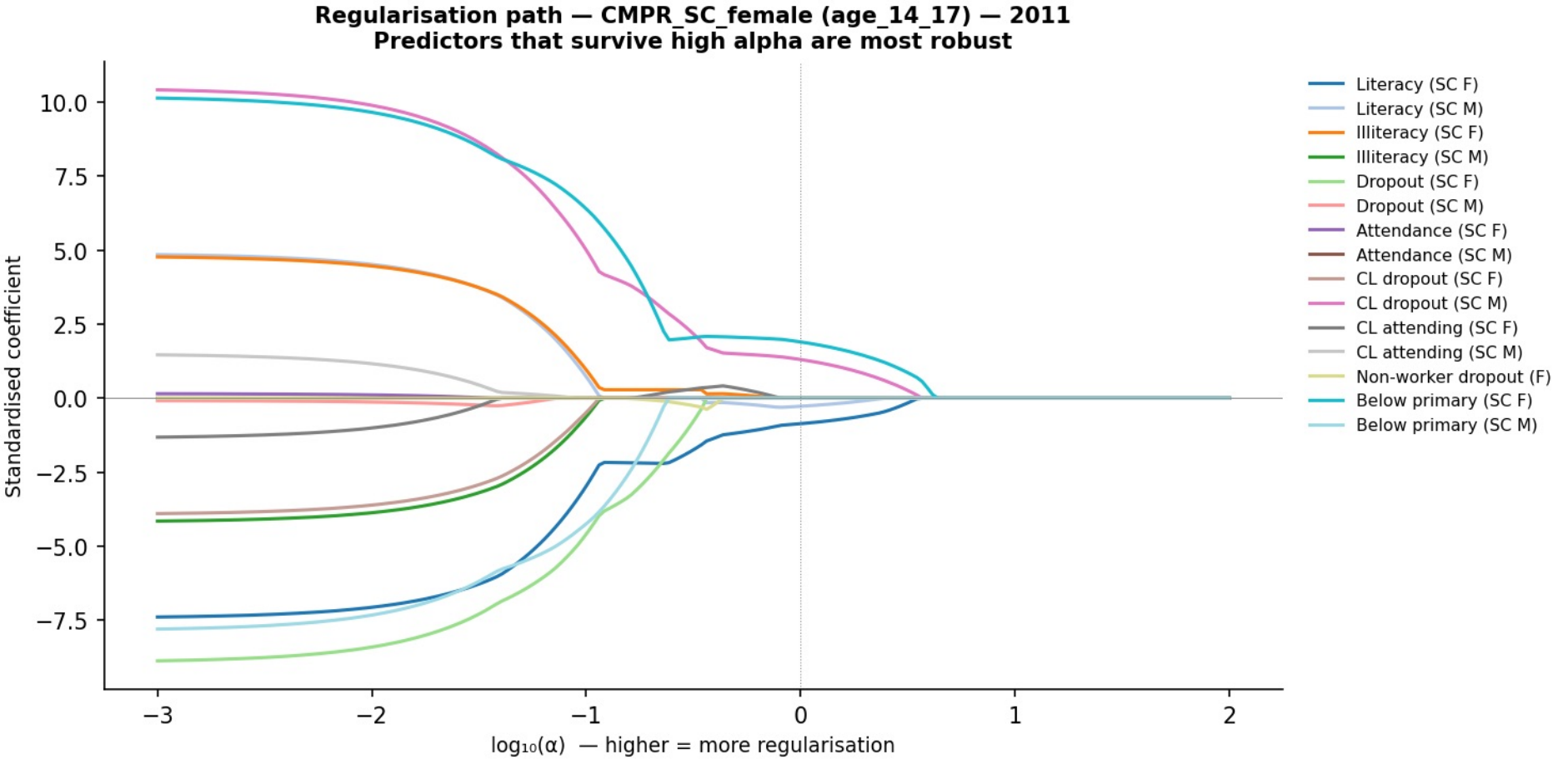

LASSO coefficients across groups — FEMALE CMPR (age_14_17)
Year: 2011 | Blue = protective | Red = risk factor | White = zeroed by LASSO
Literacy (SC F)
Literacy (SC M)
Illiteracy (SC F)
CL dropout (SC M)
Below primary (SC F)
Dropout (ST F)
Attendance (ST F)
CL dropout (ST M)
CL attending (ST M)
Non-worker dropout (F)
Below primary (ST F)
Literacy (Hindu F)
Below primary (Muslim F)
Literacy (Christian F)
Below primary (Christian F)
Literacy (Total F)
-0.72
-0.24
+0.08
+1.21
+1.81
+3.09
-0.31
+0.38
-0.92
+0.40
+1.66
-2.19
+2.71
-1.19
+1.41
-0.99
SC
ST
Hindu
Muslim
Christian
Total
Standardised coefficient
3
2
1
0
−1
−2
−3

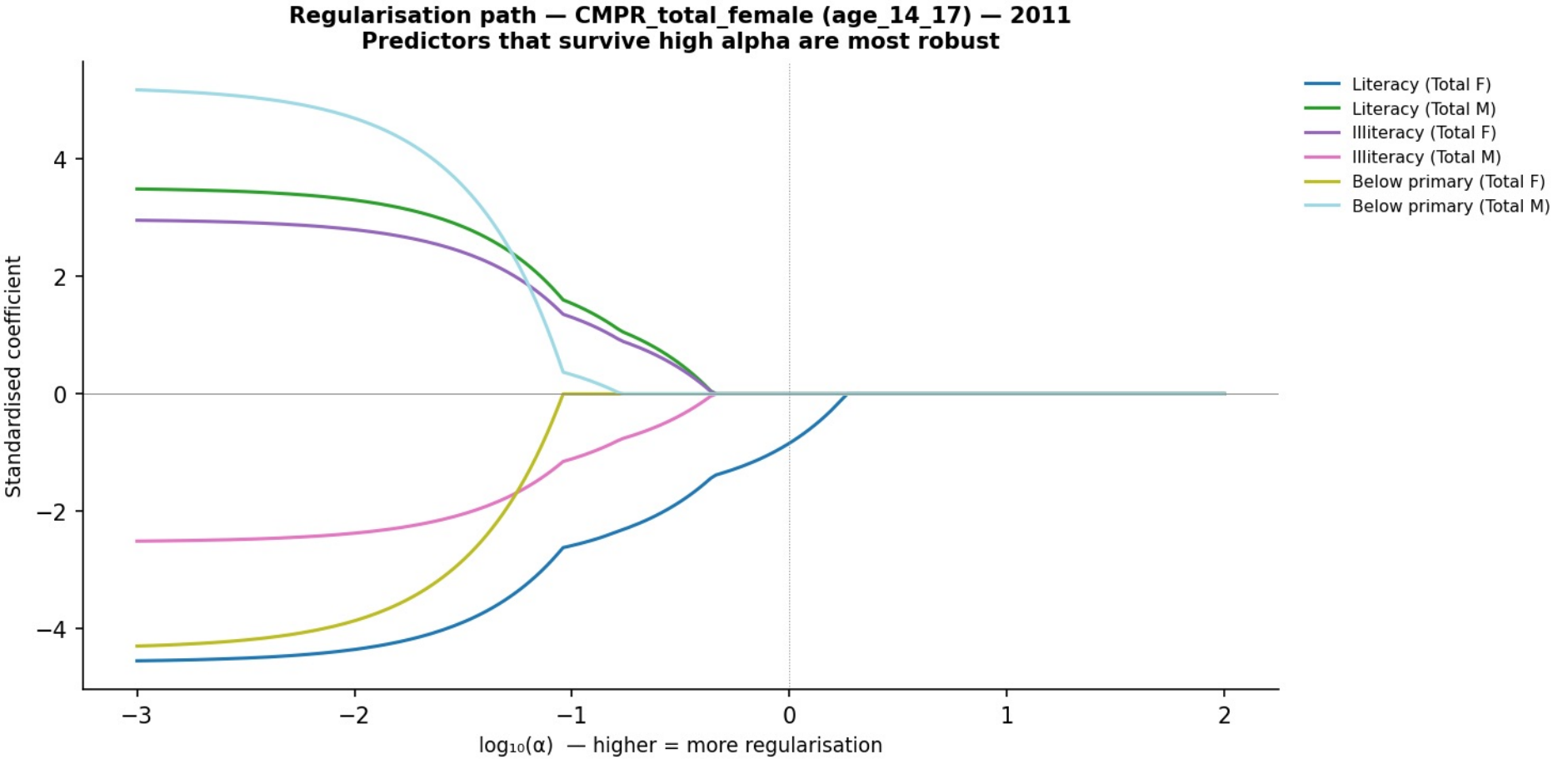

What changed between 2001 and 2011?
Target: CMPR_hindu_female | bracket: age_14_17
Middle school (Hindu F)
DROPPED 2001:-3.00
Literacy (Hindu F)
2001:-3.12 → 2011:-2.19
Literacy (Hindu F)
DROPPED 2001:+0.00
New driver in 2011
Dropped out in 2011
Persistent (coefficient changed)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Change in coefficient (2011 − 2001)

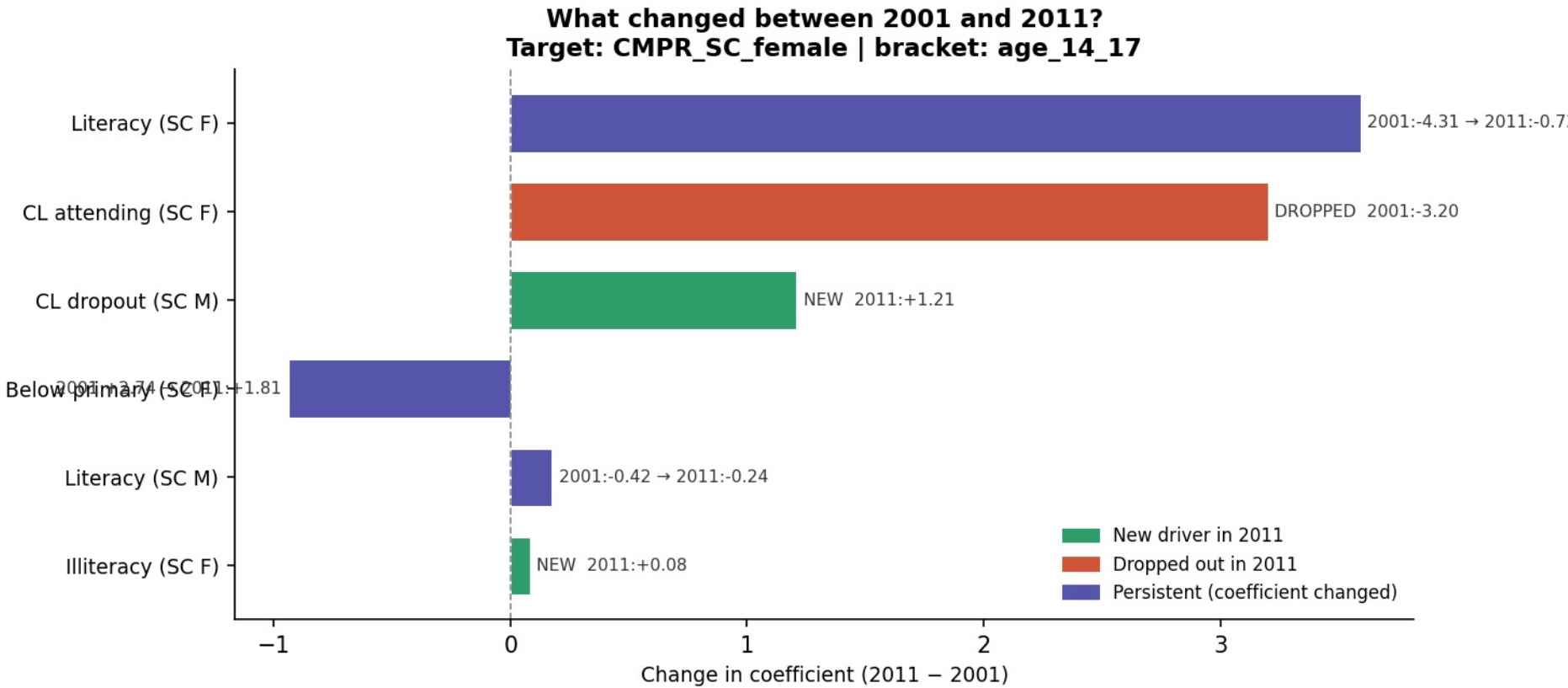

LASSO coefficients — CMPR_hindu_female (age_14_17)
Standardised predictors — bar width = effect size on CMPR
2001
α = 0.8498 | R² train = 0.45 | R² CV = 0.26 | n = 35
2011
α = 0.9545 | R² train = 0.22 | R² CV = 0.12 | n = 35
Literacy (Hindu M)
Illiteracy (Hindu F)
Illiteracy (Hindu M)
Below primary (Hindu F)
Middle school (Hindu F)
Literacy (Hindu F)
-0.00
-3.00
-3.12
-2.19
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
−3.0
Standardised coefficient
Protective (−)
Risk factor (+)
Zeroed by LASSO

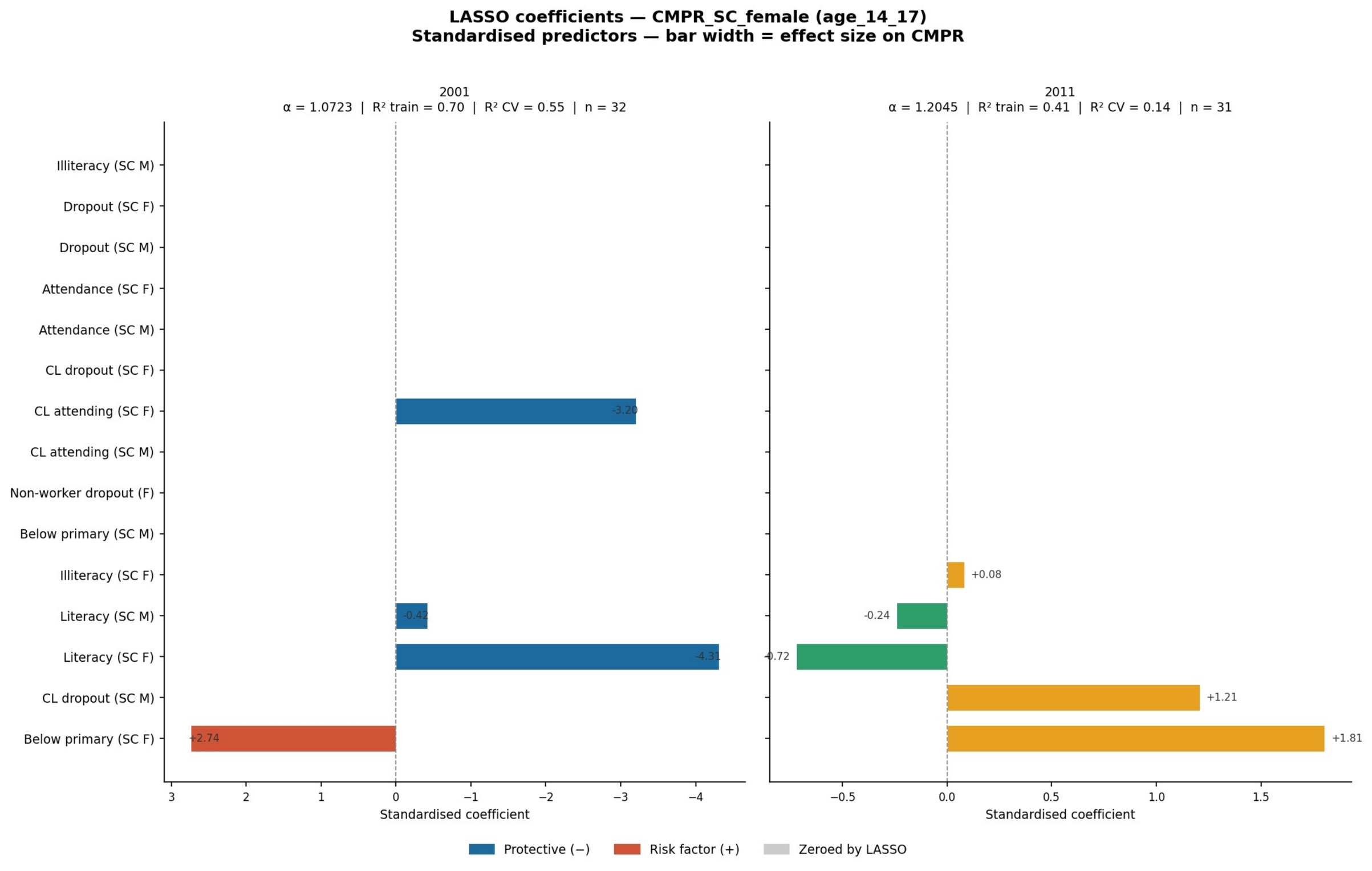

What changed between 2001 and 2011?
Target: CMPR_muslim_female | bracket: age_14_17
Literacy (Muslim F)
DROPPED 2001:-2.38
Below primary (Muslim F)
2001:+2.05 2011:+2.71
Illiteracy (Muslim F)
DROPPED 2001:+0.00
New driver in 2011
Dropped out in 2011
Persistent (coefficient changed)
0.0
0.5
1.0
1.5
2.0
2.5
Change in coefficient (2011 − 2001)

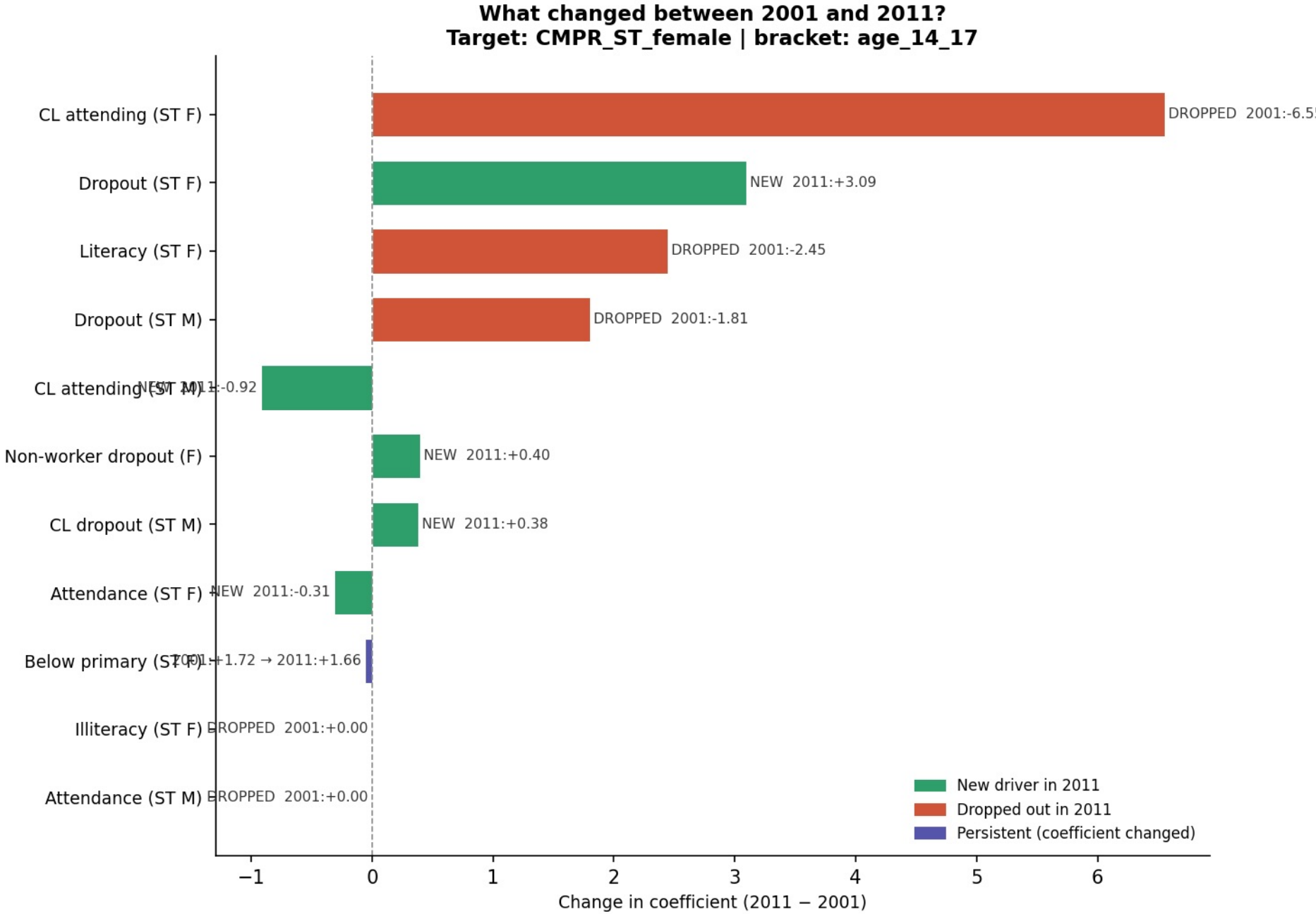

LASSO coefficients — CMPR_muslim_female (age_14_17)
Standardised predictors — bar width = effect size on CMPR
2001
α = 0.4229 | R² train = 0.47 | R² CV = 0.36 | n = 35
2011
α = 0.7565 | R² train = 0.31 | R² CV = 0.21 | n = 35
Literacy (Muslim F)
Literacy (Muslim M)
Illiteracy (Muslim F)
Illiteracy (Muslim M)
Middle school (Muslim F)
Below primary (Muslim F)
-2.38
+0.00
+3.05
+2.71
3
2
1
0
−1
−2
0.0
0.5
1.0
1.5
2.0
2.5
Standardised coefficient
Standardised coefficient
Protective (−)
Risk factor (+)
Zeroed by LASSO

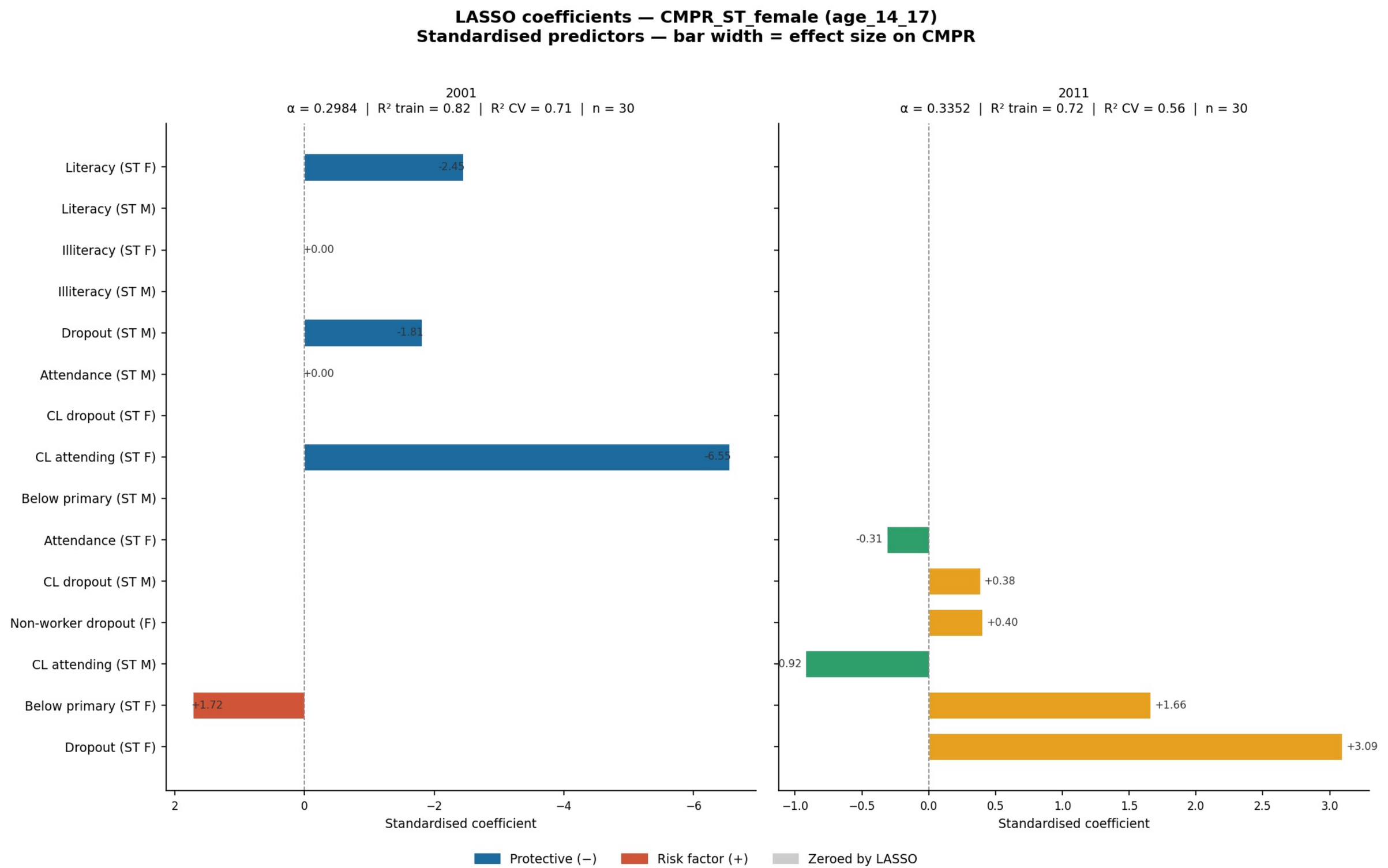

Regularisation path — CMPR_SC_female (age_14_17) — 2011
Predictors that survive high alpha are most robust
Standardised coefficient
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−3
−2
−1
0
1
2
log₁₀(α) — higher = more regularisation
Literacy (SC F)
Literacy (SC M)
Illiteracy (SC F)
Illiteracy (SC M)
Dropout (SC F)
Dropout (SC M)
Attendance (SC F)
Attendance (SC M)
CL dropout (SC F)
CL dropout (SC M)
CL attending (SC F)
CL attending (SC M)
Non-worker dropout (F)
Below primary (SC F)
Below primary (SC M)


Regularisation path — CMPR_total_female (age_14_17) — 2011
Predictors that survive high alpha are most robust
Standardised coefficient
4
2
0
−2
−4
−3
−2
−1
0
1
2
log₁₀(α) — higher = more regularisation
Literacy (Total F)
Literacy (Total M)
Illiteracy (Total F)
Illiteracy (Total M)
Below primary (Total F)
Below primary (Total M)


What changed between 2001 and 2011?
Target: CMPR_SC_female | bracket: age_14_17
Literacy (SC F)
CL attending (SC F)
CL dropout (SC M)
Below primary (SC F)
Literacy (SC M)
Illiteracy (SC F)
2001:-4.31 → 2011:-0.72
DROPPED 2001:-3.20
NEW 2011:+1.21
2001:-3.74 → 2011:+1.81
2001:-0.42 → 2011:-0.24
NEW 2011:+0.08
New driver in 2011
Dropped out in 2011
Persistent (coefficient changed)
−1
0
1
2
3
Change in coefficient (2011 − 2001)


LASSO coefficients — CMPR_SC_female (age_14_17)
Standardised predictors — bar width = effect size on CMPR
2001
α = 1.0723 | R² train = 0.70 | R² CV = 0.55 | n = 32
2011
α = 1.2045 | R² train = 0.41 | R² CV = 0.14 | n = 31
Illiteracy (SC M)
Dropout (SC F)
Dropout (SC M)
Attendance (SC F)
Attendance (SC M)
CL dropout (SC F)
CL attending (SC F)
CL attending (SC M)
Non-worker dropout (F)
Below primary (SC M)
Illiteracy (SC F)
Literacy (SC M)
Literacy (SC F)
CL dropout (SC M)
Below primary (SC F)
-3.20
-0.42
-4.31
+2.74
+0.08
-0.24
-0.72
+1.21
+1.81
3
2
1
0
−1
−2
−3
−4
−0.5
0.0
0.5
1.0
1.5
Standardised coefficient
Standardised coefficient
Protective (−)
Risk factor (+)
Zeroed by LASSO


What changed between 2001 and 2011?
Target: CMPR_ST_female | bracket: age_14_17
CL attending (ST F)
Dropout (ST F)
Literacy (ST F)
Dropout (ST M)
CL attending (ST M)
Non-worker dropout (F)
CL dropout (ST M)
Attendance (ST F)
Below primary (ST F)
Illiteracy (ST F)
Attendance (ST M)
DROPPED 2001:-6.55
NEW 2011:+3.09
DROPPED 2001:-2.45
DROPPED 2001:-1.81
NEW 2011:-0.92
NEW 2011:+0.40
NEW 2011:+0.38
NEW 2011:-0.31
2001:+1.72 → 2011:+1.66
DROPPED 2001:+0.00
DROPPED 2001:+0.00
New driver in 2011
Dropped out in 2011
Persistent (coefficient changed)
−1
0
1
2
3
4
5
6
Change in coefficient (2011 − 2001)


LASSO coefficients — CMPR_ST_female (age_14_17)
Standardised predictors — bar width = effect size on CMPR
2001
α = 0.2984 | R² train = 0.82 | R² CV = 0.71 | n = 30
2011
α = 0.3352 | R² train = 0.72 | R² CV = 0.56 | n = 30
Literacy (ST F)
Literacy (ST M)
Illiteracy (ST F)
Illiteracy (ST M)
Dropout (ST M)
Attendance (ST M)
CL dropout (ST F)
CL attending (ST F)
Below primary (ST M)
Attendance (ST F)
CL dropout (ST M)
Non-worker dropout (F)
CL attending (ST M)
Below primary (ST F)
Dropout (ST F)
-2.45
+0.00
-1.81
+0.00
-6.55
+1.72
-0.31
+0.38
+0.40
-0.92
+1.66
+3.09
2
0
−2
−4
−6
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Standardised coefficient
Standardised coefficient
Protective (−)
Risk factor (+)
Zeroed by LASSO